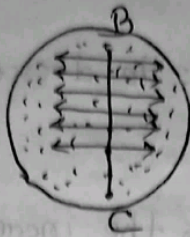


Surface tension:

If a straight line is imagined to be drawn on the surface of a liquid, the portion of the liquid of the straight line will tend to draw away from each other. But the consequent split of the liquid surface is prevented by the force of attraction betⁿ (between) the molecules of the liquid on the opposite sides of the straight line. This force of attraction per unit length of the boundary surface of the liquid is called the surface tension.

Let a straight line BC of length L be drawn on the free surface in any direction of the liquid.

Again let the force on both sides of the line perpendicular to the

$$\therefore \text{Surface tension} = \frac{\text{Force}}{\text{Length}}$$

$$\therefore T = \frac{F}{L}$$

unit:

Dimension:

Exm

→

Characteristics of surface tension:

(i) Surface tension tends to decrease surface of a liquid.

(ii) Surface tension resists the increase of area of the surface of liquid.

* Surface energy:

Dimension

Relation between surface tension and surface energy.

* Molecular theory of surface tension.

Molecular theory of surface tension:

i) Adhesive force:

ii) Cohesive force:

iii) molecular force:

iv) molecular range:

1) Force of attraction

2) Force of repulsion

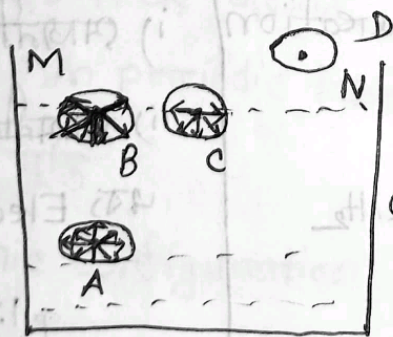
* Laplace molecular theory:

i) Sphere A is well inside the liquid

ii) B, near the surface of the liquid

iii) C, just on the surface

iv) D, above the liquid surface.



Point 1: The molecule A is totally inside the liquid. So, it will feel the cohesive force from its nearby molecules. As the force is felt from every side, the molecule will feel net 0 cohesive force.

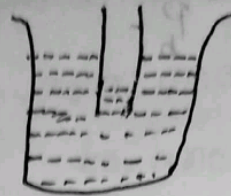
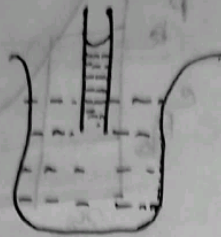
Point 2: The molecule B is partly outside and mostly inside of the liquid. So, it will feel the downward force from the nearby liquid molecules as well as the adhesive force from vapour molecules. So, a resultant downward force acting on B.

Point 3: Since the molecule C lies on the surface of the liquid the one-half of its sphere of influence lies inside the liquid and the other half of its sphere is outside the liquid. So, the cohesive force from

Conclusion:

⇒ It is thus clear that all over the surface of the liquid MN there is a downward pull due to attraction between molecules.
⇒ The surface tends to ~~contract~~ contract, i.e. it tends to decrease the area of surface MN

Angle of contact:



Capillarity:

Q. Why water rise in capillary tube and mercury fell that

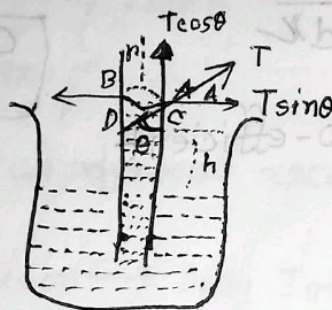


Fig 1

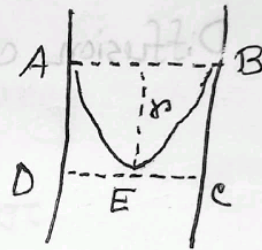


Fig-2

Capillarity:

Let us consider a capillary tube of radius " r " dipped in a water due to the surface tension the water rises inside the capillary tube. Let " h " be the height of the water column above the surface upto the lowest meniscus surface tension T acts tangentially to the surface at the point of contact. The component $T \cos \theta$ acts vertically upward

and $T \sin \theta$ acts in the horizontal direction. But $T \sin \theta$ cancels each other and only $T \cos \theta$ is effective.

$$\text{Force per unit length} = T \cos \theta$$

$$\text{Circumference} = 2\pi r$$

$$\begin{aligned} \text{Total upward force} &= T \cos \theta \times 2\pi r \\ &= 2\pi r T \cos \theta \dots (1) \end{aligned}$$

This upward force balances the weight of the liquid column in the capillary tube liquid column $= h$

Volume of the liquid in meniscus = Volume of the cylinder ABCD -

$$\begin{aligned} \text{volume of the semi-sphere, AEB} &= \pi r^2 \times r - \frac{2}{3} \pi r^3 \\ &= \pi r^3 - \frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^3 \end{aligned}$$

$$\text{volume of liquid in liquid column, } V = \pi r^2 h$$

$$\begin{aligned} \text{Total volume of liquid in column} &= V + v \\ &= (\pi r^2 h + \frac{1}{3} \pi r^3) = \pi r^2 (h + \frac{1}{3} r) \end{aligned}$$

$$\text{Density of the liquid} = \rho$$

$$\therefore \text{Weight of the liquid} = \pi r^2 (h + \frac{1}{3} r) \rho g \dots (2)$$

Now from eq (1) and (2) we can write

$$2\pi r T \cos \theta = \pi r^2 (h + \frac{1}{3} r) \rho g \quad \Rightarrow T = \frac{r \rho g (h + \frac{1}{3} r)}{2 \cos \theta} \dots (3) \quad (4)$$

$$\therefore T = \frac{\pi r^2 (h + \frac{1}{3} r) \rho g}{2\pi r \cos \theta} \dots (3)$$

If r is very small, then we can neglect $\frac{r}{3}$

$$\therefore T = \frac{r\rho gh}{2\cos\theta} \rightarrow (5)$$

$$h = \frac{2T\cos\theta}{r\rho g} \rightarrow (6)$$

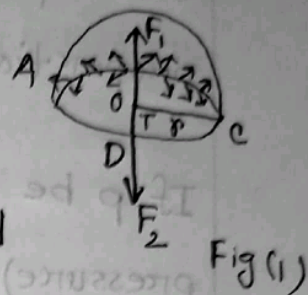
For pure water and clean glass surface $\theta = 0^\circ$
the n form eq (5)

~~$$T = \frac{r\rho g}{2} (h + \frac{r}{3}) \rightarrow (7)$$~~

By using any of the equation 4, 5, 6, 7 surface tension can be determined. By using equation 6 we can find out the height of water in the capillary tube.

Excess pressure inside a soap bubble:

In fig(1) a spherical soap bubble is seen divided along the diameter by an imaginary plane. Let the radius of the bubble be r and surface tension of soap water T . The pressure inside the bubble is greater than the pressure outside by P i.e. excess pressure P .



Now the upward force on ABCD surface $\vec{OF}_1 = \text{difference}$
in pressure $\times$ area of cross section
 $= P\pi r^2$

Again the downward force acts due to surface tension along the circumference of ABCD. This downward force acts along the point O.

Since the film has two surfaces, so force due to surface tension

$$\vec{OF}_2 = 2 \times T \times (2\pi r)$$

In equilibrium, $\vec{OF}_1 = \vec{OF}_2$

$$\Rightarrow P\pi r^2 = 2 \times T \times 2\pi r \Rightarrow P = \frac{4\pi r T}{\pi r^2}$$

$$\Rightarrow P = \frac{4T}{r} \rightarrow (i)$$

* If, T is constant, $P = \frac{1}{r} (\text{constant})$

$$\therefore P \propto \frac{1}{r} \rightarrow (ii)$$



Soap bubble



L. drop



Water bubble

Liquid

Surface

According to eq. (2), it is concluded that, the excess pressure of a soap bubble is inversely proportional to the radius of soap bubble.

If p be the outside pressure (atmospheric pressure), then the total pressure inside the bubble is $P + p$

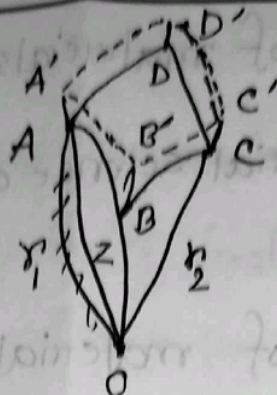
So, total pressure = $P + p$

$$= P + \frac{4T}{r}$$

In the case of liquid drop or bubble surrounded by liquid, there is only one surface so excess pressure $p = \frac{2T}{r}$

Prove that,

$$P = 2T \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$



$$AB = x$$

$$BC = y$$

$$BB' = dz$$

$$A'B' = x + dx$$

$$B'C' = y + dy$$

If we consider a curved membrane where the element of the curved membrane is in equilibrium only if an excess of pressure acts on the concave side (surface to the center) and balance the surface tension effect if we denote the excess pressure by P were done.

The work done by pressure,

$$W_p = \text{Force} \times \text{displacement}$$

$$W_p = P \times x y \times dz$$

$$= P x y dz \rightarrow (i)$$

$$[\text{Force} = \text{Pressure} \times \text{Area}]$$

$$F = P \times A$$

Work done to increase the length of one side of the membrane through dx is given by

$$W_{Tx} = T_y dx$$

Similarly, $W_{Ty} = T_x dy$

Total work done against surface tension force,

$$\begin{aligned} W_T &= W_{Tx} + W_{Ty} \\ &= T_y dx + T_x dy \\ &= T(y dx + x dy) \rightarrow (2) \end{aligned}$$

In the triangle ABO and A'B'O

$$\frac{A'B'}{AB} = \frac{A'O}{AO}$$

or, $\frac{x+dx}{x} = \frac{y+dy}{y}$

or, $1 + \frac{dx}{x} = 1 + \frac{dy}{y}$

or, $\frac{1}{x} = \frac{1}{y} \frac{dy}{dx}$

or, $\frac{1}{y} = \frac{1}{x} \frac{dx}{dy} \rightarrow (3)$

Similarly,

$$\frac{1}{y} = \frac{1}{r_2} \frac{dz}{dy}$$

$$\Rightarrow \frac{1}{r_2} = \frac{1}{y} \frac{dy}{dz} \rightarrow (4)$$

At equilibrium,

$$W_p = W_T$$

$$\therefore Pxydz = T(xdy + ydx)$$

$$\text{or, } Pxydz = Txy \left(\frac{1}{y} dy + \frac{1}{x} dx \right)$$

$$\text{or, } Pdz = T \left(\frac{dy}{y} + \frac{dx}{x} \right)$$

$$\text{or, } Pdz = Tdy \left(\frac{dz}{y} \right) + Tdx \left(\frac{dz}{x} \right)$$

$$\text{or, } P = T \left\{ \frac{1}{r_1} + \frac{1}{r_2} \right\}$$

If there are two surfaces (In case of soap bubble)

$$P = 2T \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

(ii) In case of mercury drop,

$$P = T \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

(iii) In case of ~~mer~~ air bubble, inside a liquid

$$P = T \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

Water, Mercury bubble
→ 1 surface

Air bubble → 2 surfaces

iv) If the surface is spherical

$$r_1 = r_2 = r$$

$$P = 2T \left(\frac{1}{r_1} + \frac{1}{r_2} \right) = 2T \left(\frac{1}{r} + \frac{1}{r} \right)$$

$$= \frac{2T \times 2}{r} = \frac{4T}{r}$$

$$\therefore P = \frac{4T}{r}$$

v) If the surface is cylindrical surface,

$$r_2 = \infty \Rightarrow \frac{1}{r_2} = 0$$

$$\therefore P = T \left(\frac{1}{r} + 0 \right)$$

$$\therefore P = \frac{T}{r}$$